

Winter chill for apricot in Spain — the pipeline on one page

Where does **'Búlida'** (47.5 chill portions required) stop meeting its requirement while its somatic mutant **'Búlida Precoz'** (33.7) still meets its own, and how does that band of land move through the century? Chill is computed at 3,460 weather stations from 11 CMIP6 models, interpolated to a 1 km grid, and reported as **area of cropland** rather than as a count of stations. Every number below is read from the project's canonical tables when this sheet is built.

1 · THE FOUR DATA SOURCES

PROJECTIONS

PNACC AR6 - ESD-RegBA at stations

AdapteCCa THREDDS, direct HTTP, no credential

product group SP-005

files	88 NetCDF, ~15 GB
shape	11 models × 4 experiments × 2 variables
stations	3,460
span	historical 1950–2014, SSP 2015–2100
layout	[time × station], days since 1850

The portal's web form serves the same product over 3,044 stations, not 3,460. The route has to be stated in Methods or the areas cannot be reproduced.

OBSERVED

PNACC observational archive

web form, delivered by email

not available as daily data on THREDDS

stations	3,044
span	1975-01-01 to 2020-12-31
days	16,802 continuous
layout	[station × time], hours since 1900
ends	2020, and cannot be extended by this route

RECENT

AEMET OpenData, daily climatological values

REST API, two-step (envelope then payload)

key read from \$AEMET_API_KEY; never written to disk

stations	666 of the 3,044
span	1995–2025
limit	6 months per request, ~62 per station
rate	40 calls/min against a ~50 ceiling
role	extension only; never overwrites the archive

Covers 22% of the network and only 131 stations reach back to 1995, so it is a cross-check and an extension, not a baseline.

LAND

CORINE Land Cover 2018

Copernicus raster, 100 m

EPSG:3035, 46,000 × 65,000 cells

classes	211–244, excluding 231 pasture
area	229,676 km²
share	46% of Spain
use	denominator of every area figure
weighting	cropland fraction per 1 km cell

2 · WHERE EACH STEP RUNS

HPC cluster (Ladon) — the heavy end		
Download 14_ladon_download_thredds.sh 88 files by wget, skipped if already non-empty. No OPeNDAP, so whole files come down and are subset locally.	Chill engine 15_chill_national_parallel.R Reads NetCDF directly, clips to the window, parallelises by station, checkpoints per model×scenario with atomic writes. ~23 h for the full run.	Recent observed 21_aemet_observed_download.py Standard library only, so it runs under any Python on the cluster. Resumable, and refuses to resume a directory holding a different year range.

Local workstation — surfaces, validation and figures				
Merge 22_merge_chill_tables.R One canonical table, 462,808 rows, labelled by period and class. Refuses input whose coordinates have been through Excel.	Surface 19_cropland_viability_national.R Median across models, IDW, CORINE mask, three classes, area weighted by cropland fraction.	Per model 36_per_model_stats.R The same chain repeated once per model, so the spread survives instead of being collapsed. Feeds the agreement hatching.	Validation 41_idw_crossval.R Leave-one-out over the interpolation, with the same parameters. Two self-checks must pass before anything is written.	Figures 33, 34, 37, 38, 42, 43 Every number read from the canonical tables at build time. Nothing typed.

3 · PARAMETERS, AND WHERE EACH ONE IS SET

Chill model	Dynamic Model, Fishman et al. (1987) parametrisation	DM_JOSE.R:4-5
Model constants	E0 4457.8 · E1 10161.9 · A0 419700 · A1 1.797×10 ¹⁴ · slope 1.6 · Tf 277	DM_JOSE.R:4-5
Chill season	Julian day 305 to 59 (1 Nov to 28 Feb)	15_chill_national_parallel.R:116
Safe Winter Chill	10th percentile of seasonal chill portions, across seasons within a station	15_chill_national_parallel.R:346
Season kept if	at least 85% of days present	15_chill_national_parallel.R:116, :339
Station kept if	no more than 40% missing in Tmax or Tmin, and 3 or more valid seasons	15_chill_national_parallel.R:117, :344
Fill-value guard	values outside –90 to 70 °C masked (four models ship –999 undeclared)	15_chill_national_parallel.R:118, :271
Baseline splice	historical to 2014, then SSP2-4.5 from 2015 (CMIP6 historical ends 2014 by design)	15_chill_national_parallel.R:159, :295-302
Time windows	1995–2020 baseline, then 2021–2040, 2041–2070, 2071–2100 (tiling, no overlap)	15_chill_national_parallel.R:126-132
Ensemble statistic	median across the 11 models, taken at the station, before interpolation	19_cropland_viability_national.R:66
Interpolation	IDW, power 2, 50 km radius, at most 12 neighbours; the radius is the mask	19_cropland_viability_national.R:43-45, :151
Grid	1 km, EPSG:3035 (equal-area); cell area from the realised resolution, not the nominal one	19_cropland_viability_national.R:37-40; 00_corine.R:43
Cropland	CORINE 2018, 100 m, classes 211–244 excluding 231 pasture; each cell weighted by its cropland fraction	00_corine.R:24-28; 19_...:124
Cultivar thresholds	'Búlida' 47.5 CP, 'Búlida Precoz' 33.7 CP (Ruiz et al. 2019, both ±3.3)	19_cropland_viability_national.R:41-42
Model agreement	hatched where fewer than 9 of 11 models agree on the class (AR6 80% convention)	00_hatch.R:28, :107-113

How the Dynamic Model enters the chain. DM_JOSE.R carries the Fishman et al. (1987) parametrisation and is used unmodified: the function object is passed straight into chillR as one of the models, with Utah computed alongside it as a secondary metric. The daily-to-hourly reconstruction happens inside chillR from the station's latitude.

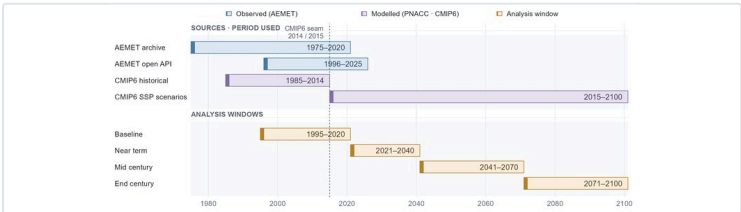
```
weather <- fix_weather(df, end_at_present = FALSE)  
  
tempResponse_daily_list(list(weather), latitude = lat,  
                          Start_JDay = 305, End_JDay = 59,  
                          models = list(Utah_Chill_Units = Utah_Model,  
                                         Chill_Portions = DM_JOSE))
```

Why the parametrisation matters. chillR's default Dynamic_Model carries the 1988 constants and gives different numbers: measured over 14 seasons at Cieza, 6.94 CP lower on average. The two cultivars are separated by 13.8 CP, so supply and demand have to be on the same scale or the comparison means nothing.

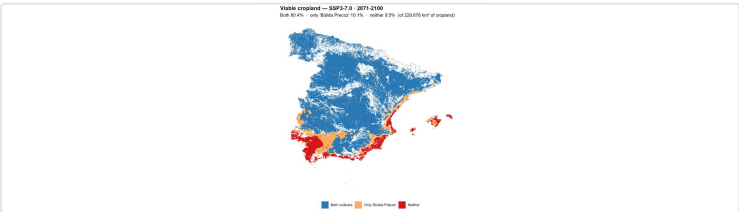
What the ensemble hides. The maps show the median of 11 models, but the models disagree by 21.3 CP at end of century against 2.9 CP at the baseline. UKESM1-0-LL leaves 41.1 CP and IITM-ESM leaves 62.3. On area, the land where neither cultivar works ranges from 4,983 to 75,951 km² and the rescued fraction from 39.1% to 82.1%, depending on the model. 22.7% of cropland sits within one interpolation RMSE of a threshold.

4 · WHAT COMES OUT

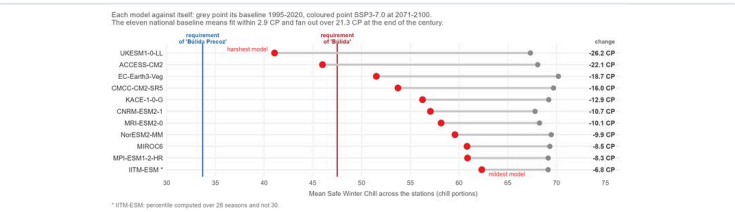
229,676 km² Spanish cropland, the denominator	45,103 km² where 'Búlida' fails by 2071–2100, SSP3-7.0	23,310 km² of that, still viable for the mutant (51.7%)	21.3 CP spread between models, against a 13.8 CP cultivar gap	6.44 CP interpolation error at 2071–2100, 46.7% of that gap
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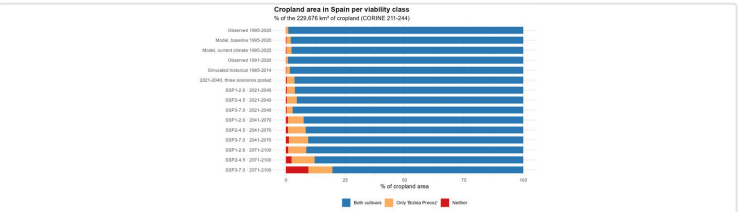
What each source covers, and which stretch of it is used



Where each cultivar still works, SSP3-7.0 at 2071–2100



The eleven models ranked by the chill they leave



The same three classes across every scenario and window